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Physical analysis of barrier-inhomogeneity effects on the electrical parameters of GaN Schottky barrier diodes

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E-mail: rogerantonio@usal.es**Keywords:** GaN, barrier inhomogeneity, Schottky barrier diode, Monte Carlo simulator

Abstract

The influence on current–voltage and capacitance–voltage characteristics of different inhomogeneous profiles for the energy barrier (ϕ_B) along the Schottky contact of GaN Schottky barrier diodes is studied by means of a home-made Monte Carlo simulator. An insight analysis of the electron concentration and electric field inside the device allows explaining the strong influence of inhomogeneities on the electronic transport across the metal semiconductor interface. In reverse bias, the low-barrier regions increase the number of tunneled electrons and these carriers tend to move towards the surrounded low populated zones inside the semiconductor. Therefore, two-dimensional effects appear and the current flowing through the device depends not only on the value and area of the low-barrier regions but also on its relative location. On the contrary, the potential profile is not affected much by low-barrier inhomogeneities, since they are pinched-off by the surrounding high-barrier zones. As a consequence, the effective ϕ_B obtained from the C – V characteristics essentially corresponds to the (high-barrier) nominal ϕ_B , while the effective ϕ_B obtained from forward I – V characteristics is strongly impacted by the low-barrier regions. In the latter case, lower effective values of the barrier are obtained together with an increase in the ideality factor. These effects connect the physical performance of these devices with the effective ϕ_B measured in fabricated ones.

1. Introduction

The high level of leakage current exhibited by GaN Schottky barrier diodes (SBDs) in reverse bias conditions [1–6] still remains a key concern for the practical use of these otherwise promising devices for high power and high frequency applications [7–9], such as consumer electronic [10, 11], renewable energy [10, 11], electric vehicle industry [10, 12], or satellite communications [13–15]. The leakage current in excess of the ideal thermionic emission and tunneling contributions are originated by several effects related to defects at the metal/semiconductor interface, leading to trap assisted tunneling, Poole–Frenkel emission, etc [1, 16]. The reduction in the Schottky barrier height (ϕ_B) typically found when estimated from the forward bias I – V curves with respect to the value extracted from C – V measurements is often attributed to inhomogeneities in ϕ_B , which may also affect the value of the leakage current [2, 3, 17–19]. This subject deserves a deep physical understanding in order to profit from GaN technology.

In this work, the influence of an inhomogeneous ϕ_B on the physical performance of GaN SBDs is analyzed by means of a 2D Monte Carlo (MC) simulator self-consistently coupled with a 2D Poisson solver [20]. Different types of inhomogeneous ϕ_B distributions are considered, mimicking the experimental values analyzed in the literature [16, 21–23]. A rectangular diode geometry consisting of Schottky and ohmic contacts edging a low-doped GaN epilayer and a highly doped substrate, respectively, together with tunnel current in reverse bias modeled by the WKB approximation [24] are considered

in order to straightforwardly understand the physical implications of inhomogeneous barriers. Since the MC simulator updates the potential profile at every time step in agreement with the charge distributions, it naturally accounts for phenomena such as the possible barrier pinch-off, i.e. any low barrier patch could be pinched-off by the surrounding high-barrier background, if its size is smaller than the Debye length [3, 23]. The model provides the two-dimensional profiles of electron concentration and electric field, offering an interesting insight into the microscopic behavior of the tunneled carriers in reverse bias conditions. The effective ϕ_B (ϕ_B^{eff}) obtained from I - V and C - V is also analyzed to connect the physical performance of the device with the experimental characteristics.

The article is organized as follows. In section 2, we include an overview of the MC simulator and the physical model for the Schottky barrier. In section 3, the results of the simulations are discussed to explain the influence of the inhomogeneities and their location. Finally, in section 4 the main conclusions of our work are summarized.

2. Physical model and simulated structures

A home-made semiclassical ensemble MC simulator self-consistently coupled with a 2-D Poisson solver [25, 26] has been employed in order to understand the physical influence of different inhomogeneous profiles of the energy barrier ϕ_B at the Schottky contact on the electronic transport across metal-semiconductor interfaces. The GaN conduction band is considered as three non-parabolic spherical valleys: Γ_1 , U, and Γ_3 . Ionized impurities, dislocations, polar and nonpolar optical phonons, acoustic phonons, piezoelectric and intervalley scattering mechanisms are included in the simulator. To ensure the accuracy of the simulations, the electric field is updated at every time step $\Delta t = 0.2$ fs and the cell size is in the range 1–5 nm, smaller than the Debye length in GaN for the considered doping levels. In particular, for the epitaxial-layer doping $N_D = 3 \times 10^{23} \text{ m}^{-3}$, $L_D = \sqrt{\epsilon_{\text{SC}} k_B T / e^2 N_D} \approx 6.5$ nm at $T = 300$ K [20], where $\epsilon_{\text{SC}} = 8.9\epsilon_0$ is the low-frequency permittivity for GaN, e is the fundamental charge, and k_B is the Boltzmann constant.

The simulated SBD structure is sketched in figure 1(S0). It consists of a 200 nm-length Schottky contact on a 350 nm-thick active layer with the mentioned doping level, followed by a 150 nm-thick substrate with $N_D^+ = 5 \times 10^{24} \text{ m}^{-3}$ [6, 25]. This simple schema allows us to focus on the contribution to the physical performance of the barrier profile. To this end, different inhomogeneous distributions of ϕ_B are considered, as shown in figure 1. Barrier profiles under analysis are similar to those found in experimental devices. Firstly, we consider a Gaussian distribution, similar to that characterized in the fabricated Au/Pt/GaN Schottky contacts of [21], which proposes that surface defects can be associated with threading dislocations. From a technological perspective, it is useful to characterize an SBD with a mean value of ϕ_B and a standard deviation σ [17]. Furthermore, the Gaussian barrier distribution is the one used in some classical models [22, 23, 27]. In [21], the measured Gaussian distribution exhibits a standard deviation between 0.11 and 0.27 eV. Thus, for our calculations, we have chosen a ϕ_B with a mean value of 0.5 eV and a standard deviation of $\sigma = 0.2$ eV (figure 1(S2)).

A second set of studied inhomogeneous ϕ_B is portrayed as isolated low-barrier regions embedded into a uniform higher background barrier [16, 22, 23]. These barrier distributions are used in Tung's models to discuss the electrical results given by Au/Pt/GaN Schottky contacts [21] and Ni/Au GaN diodes [3], providing quantitative information on the degree of spatial uniformity at the nanoscale. Accordingly, having a background reference of $\phi_B = 0.5$ eV, three different situations have been implemented as follows: including a central 0.1 eV region of 5 nm (figure 1(S3)), three 0.1 eV symmetrically-placed regions of 5 nm (figure 1(S4)), and one central low-barrier region of 15 nm (figure 1(S5)).

Additionally, to provide a qualitative but clearer comprehension of the physical influence of the inhomogeneities on ϕ_B , an inhomogeneous ϕ_B designed by dividing the SBD into three different regions is considered: 65 nm with 0.5 eV, 65 nm with 0.4 eV and 70 nm with 0.6 eV (figure 1(S6)). Note that also an homogeneous value of $\phi_B = 0.5$ eV has been taken as a reference (figure 1(S1)). Table 1 summarizes the different ϕ_B distributions described above.

In order to link the experimental measurements with the microscopic analysis, we obtain from simulations the I - V and C - V curves to extract from them the ϕ_B^{eff} and the ideality factor η , typical parameters used to characterize the SBDs.

To determine electron injection at the Schottky contact, the simulator implements a model that includes both thermionic (I_{th}) and tunneling (I_{TC}) current components. The same set of parameters can be used for both polarities, forward and reverse. For the forward bias current, only the thermionic emission is considered, whereas for the reverse current the tunneling contribution is the one that dominates the injection, although thermionic model is also included [26]. The general expression of the current density around a given energy ϵ in a $d\epsilon$ energy interval is described by $I(\epsilon)d\epsilon = -eN_{\text{m},s}(\epsilon)T_C(\epsilon)d\epsilon$,

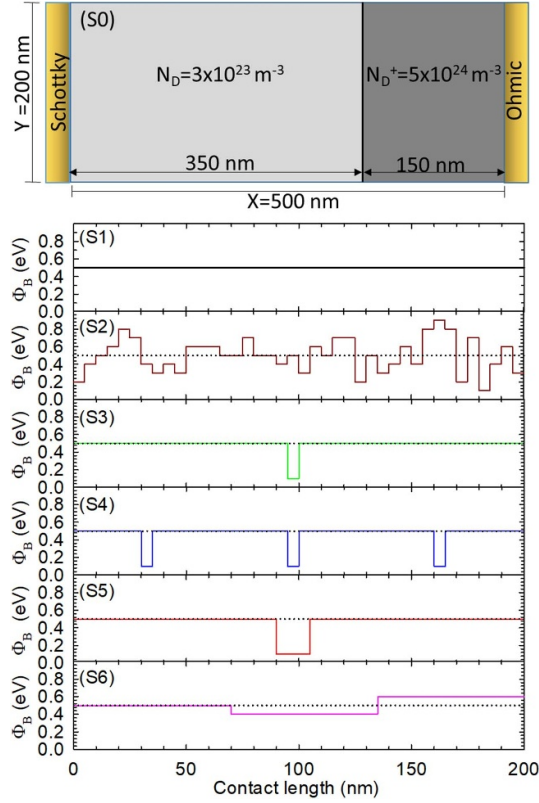


Figure 1. (S0) General structure of the simulated devices where the ϕ_B distributions can be: (S1) homogeneous ϕ_B distribution, (S2) Gaussian distribution to resemble the effect of threading dislocations [21]; a background Φ_B of 0.5 eV with a 5 nm (S3) or 15 nm (S5) central region with $\Phi_B = 0.1$ eV, (S4) three 1 nm-length regions where the value of Φ_B is reduced to 0.1 eV to resemble the effect of punctual defects on the metal-semiconductor interface [3, 21], and (S6) three regions of similar length with $\phi_B = 0.5, 0.4$ and 0.6 eV, respectively, ideally designed in order to compare its physical behavior with isolated SBDs with such ϕ_B .

Table 1. ϕ_B distributions of the simulated structures.

Barrier distribution name	ϕ_B distribution
S1	Homogeneous $\phi_B = 0.5$ eV
S2	Gaussian distribution
S3	5 nm region with $\phi_B = 0.1$ eV
S4	Three 5 nm regions with $\phi_B = 0.1$ eV
S5	15 nm region with $\phi_B = 0.1$ eV
S6	3 zones with $\phi_B = 0.5, 0.4$ and 0.6 eV

where $N_{m,s}(\epsilon)$ is the rate of incident electrons from the metal to the semiconductor and $T_C(\epsilon)$ is the transmission coefficient, both at energy ϵ . Both thermionic emission and tunneling are obtained by integrating $I(\epsilon)$ over the electron energy interval susceptible to contribute to the respective current. To estimate the transmission coefficient for the I_{TC} , we use the Wentzel–Kramer–Brillouin approximation as $T_C(\epsilon) = \exp[-2/\hbar \int_{x_1}^{x_2} (2m^*(\epsilon_c(x) - \epsilon))^{1/2} dx]$, where x_1 and x_2 are the classical turning points, $\hbar$ is the reduced Planck constant, $m^* = 0.22m_0$ is the electron effective mass and $\epsilon_c(x)$ is the energy barrier profile as obtained from the solution of the Poisson equation in MC simulations. For the thermionic emission, the transmission coefficient is 1. An exhaustive description of the model can be found in [26]. With the aim of discerning only the influence of an inhomogeneous barrier, image force lowering or defect-related effects such as trap-assisted tunneling, Poole Frenkel, or interfacial dipoles emerging due to the fabrication process [28], are not included, allowing to precise what characteristic changes are due to the different inhomogeneous distributions.

In this context, the ϕ_B^{eff} is calculated from the forward I – V curves by comparing the numeric results with those obtained by means of a simple thermionic emission model:

$$I_{\text{th}} = I_0 \exp\left(\frac{e(V - I_{\text{th}}R_s)}{\eta k_B T}\right), \quad (1)$$

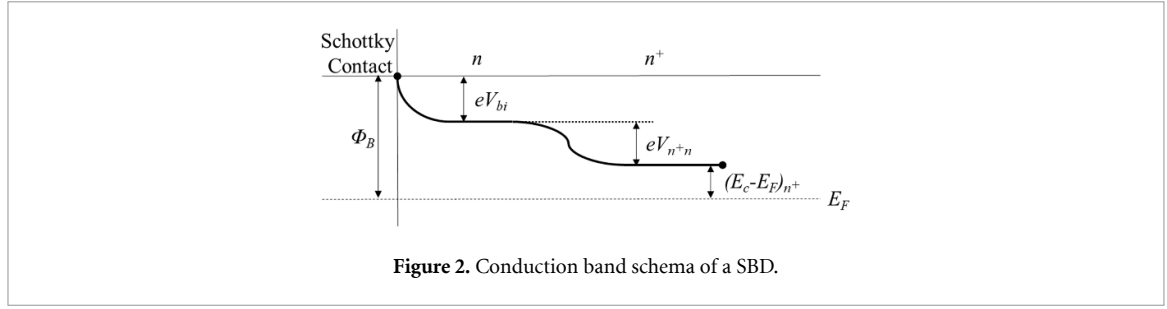


Figure 2. Conduction band schema of a SBD.

where T is the absolute temperature, V is the applied voltage, R_s is the series resistance arising intrinsically in the simulations from electron transport in the GaN bulk region and

$$I_0 = A^* T^2 \exp\left(\frac{-\phi_B}{k_B T}\right), \quad (2)$$

being $A^* = 4\pi m^* k_B e / h^3$ the effective Richardson constant and h , the Planck constant.

It is also possible to extract the ϕ_B^{eff} from the C - V simulated curves. In this case, ϕ_B is obtained as a result of the addition of the parameters described in figure 2, given by the following equation :

$$\phi_B = eV_{bi} + eV_{n+n} + (E_c - E_F)_{n+} \quad (3)$$

where e is the electron charge, V_{bi} is the built-in voltage, $V_{n+n} = k_B T \ln \frac{N_D^+}{N_D}$ is the voltage drop between the two semiconductor layers (doped and highly doped) and $(E_c - E_F)_{n+} = -k_B T \ln \frac{N_C^+}{N_C}$, with $N_C = 2 \left(\frac{2\pi m^* k_B T}{h^2} \right)^{3/2}$ the effective density of states of the GaN conduction-band. V_{bi} is extracted from simulated C - V curves for low reverse voltages where the diode exhibits a close to ideal behavior given by

$$\frac{1}{C_{id}^2} = \frac{1}{L_{Sch}^2} \frac{2(V_{bi} - V)}{eN_D \epsilon_{SC}} \quad (4)$$

where C_{id} is capacitance per unit length (evoking the non-simulated dimension in the 2D MC model), L_{Sch} is the length of the Schottky contact and V the applied voltage.

3. Results and discussion

Firstly, the reverse I - V characteristics of the SBD under study with the different ϕ_B distributions are plotted in figure 3. The reference case considering an homogeneous $\phi_B = 0.5$ eV (S1) is plotted for comparison. All structures provide higher currents when any type of inhomogeneity in ϕ_B is considered: local low-barrier regions lead to higher levels of reverse current. In particular, the device with just 5 nm of $\phi_B = 0.1$ eV at the middle of the contact (S3) provides more current than that with an homogeneous ϕ_B , but not as much as the structure with a Gaussian distribution (S2), which includes more numerous regions with $\phi_B < 0.5$ eV. Even higher values of current are provided by diodes with barrier profiles (S4) and (S5), both of them with a global width of 15 nm with $\phi_B = 0.1$ eV. Thus, the reverse current is strongly affected by the lowest value of ϕ_B (in the Gaussian distribution, other values of ϕ_B in the range from 0.5 eV to 0.1 eV are considered) and the length of these low-barrier regions. It is remarkable that the discrepancy between the current provided by S5 (15 nm with $\phi_B = 0.1$ eV) and that by S4 (three 5 nm-regions with $\phi_B = 0.1$ eV). For S5 the current is higher until the high reverse voltage $V = -60$ V. This result indicates that not only the value and the length of the local low ϕ_B is important, but also the relative location of the low-barrier regions influences the resulting reverse current level. To illustrate this assertion, figure 3 also presents the reverse I - V curve for the ϕ_B distribution corresponding to S6 as well as that obtained as the total current provided by three independent SBDs with homogeneous barrier and contact length imitating each region of S6 (i.e. a 65 nm-length SBD with $\phi_B = 0.5$ eV, a 65 nm-length SBD with $\phi_B = 0.4$ eV and a 70 nm-length with $\phi_B = 0.6$ eV). The current from S6 is slightly higher than the addition of the three individual SBDs with similar overall profile. This fact proves the existence of an interaction, 2D effects, between the different ϕ_B zones that affects the physical behavior of the device.

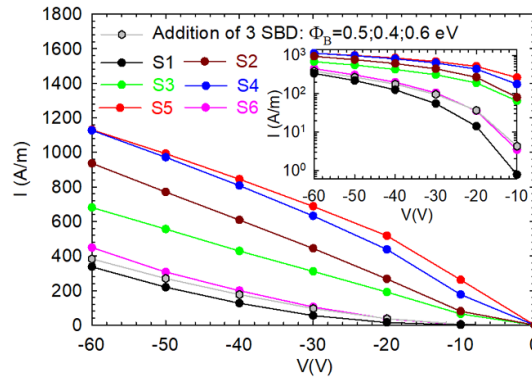


Figure 3. Reverse-bias I - V simulated characteristics in linear and logarithmic (inset) scale for SBDs with different ϕ_B distributions: (a) S1 homogeneous $\phi_B = 0.5$ eV, S2, S3, S4, S5, S6 barrier profiles, and the one extracted from adding the current of three independent SBDs corresponding to the same regions and dimensions of S6.

With the purpose of providing a deep insight into the physical performance of the devices, the electron concentrations and the component of the electric field parallel to the Schottky contact (E_y) provided by our MC simulator have been plotted in figures 4(a) and (b), respectively, for all the structures with inhomogeneous ϕ_B under analysis and an applied voltage $V = -30$ V. As expected, in all cases, the regions dominating the electron injection are those with lower ϕ_B values due to the higher contribution of the tunneling current, while the electron concentration decreases around the regions with higher value of ϕ_B .

Focusing on the characteristics of S5, in figure 4(S5(a)) it can be observed that there is a high electron concentration in the 15 nm region of the Schottky contact with $\phi_B = 0.1$ eV. The low barrier increases the tunneling probability, resulting in a higher electron injection. Note that once injected, after moving a few nanometers inside the semiconductor (around 30 nm in this case), the electrons shift up and down. As a consequence, the area of high carrier concentration also expands to the regions where the barrier is higher ($\phi_B = 0.5$ eV).

A deeper explanation of this behavior can be given when the component of the electric field E_y inside the device is analyzed, as in figure 4(S5(b)). This figure shows two different situations. Near the Schottky contact (SC) two strong field areas with E_{y1} pointing to the higher ϕ_B regions are found (inset). These fields are induced by the different potential distributions along the Schottky interface due to the different barrier heights. Both E_{y1} push the electrons inside the semiconductor into the low-barrier region, and this area keeps the high electron concentration.

The other situation is observed after a few nanometers from the SC inside the semiconductor (in this case, starting from around 20–30 nm). Now, new strong electric fields appear, E_{y2} , but in the opposite direction to the previous ones. These E_{y2} are originated by the positive ionized donor impurities of the depletion region and the very high electron concentration in the center of the structure. But in this case, these electric fields favor the presence of carriers in the regions where ϕ_B is higher. This is, the electric fields point to the low-barrier region, helping the shift of the electrons to the depleted high-barrier areas, spreading the zone of high carrier concentration. As a consequence, strong two-dimensional effects appear in the device that affect the dynamics of carriers and therefore the I - V characteristics of the diode.

The strength of the two-dimensional effects depends not only on the value of the barrier but also on the length of the low ϕ_B region. Thus, in figure 4(S3(a)), lower electron concentration inside the device is obtained. Although the same effects as those found in figure 4(S5(a)) are present, due to the fact that high carrier injection happens in 5 nm instead of 15 nm, the tunneling and the subsequent spreading of electrons and the electric fields are not as pronounced as in figure 4(S5(a)).

However, the location of the low barrier region is also important. As observed from figure 4(S4(a)), for S4 barrier profile the regions with $\phi_B = 0.1$ eV still dominate the injection. However, since there are three symmetrically-placed regions of 5 nm with $\phi_B = 0.1$ eV, there is not as much shift of the electrons as for S5(a). The values of E_{y2} are much lower than in the case of figure 4(S5(a)), when there is only one region of 15 nm. This explains the higher current achieved with S5.

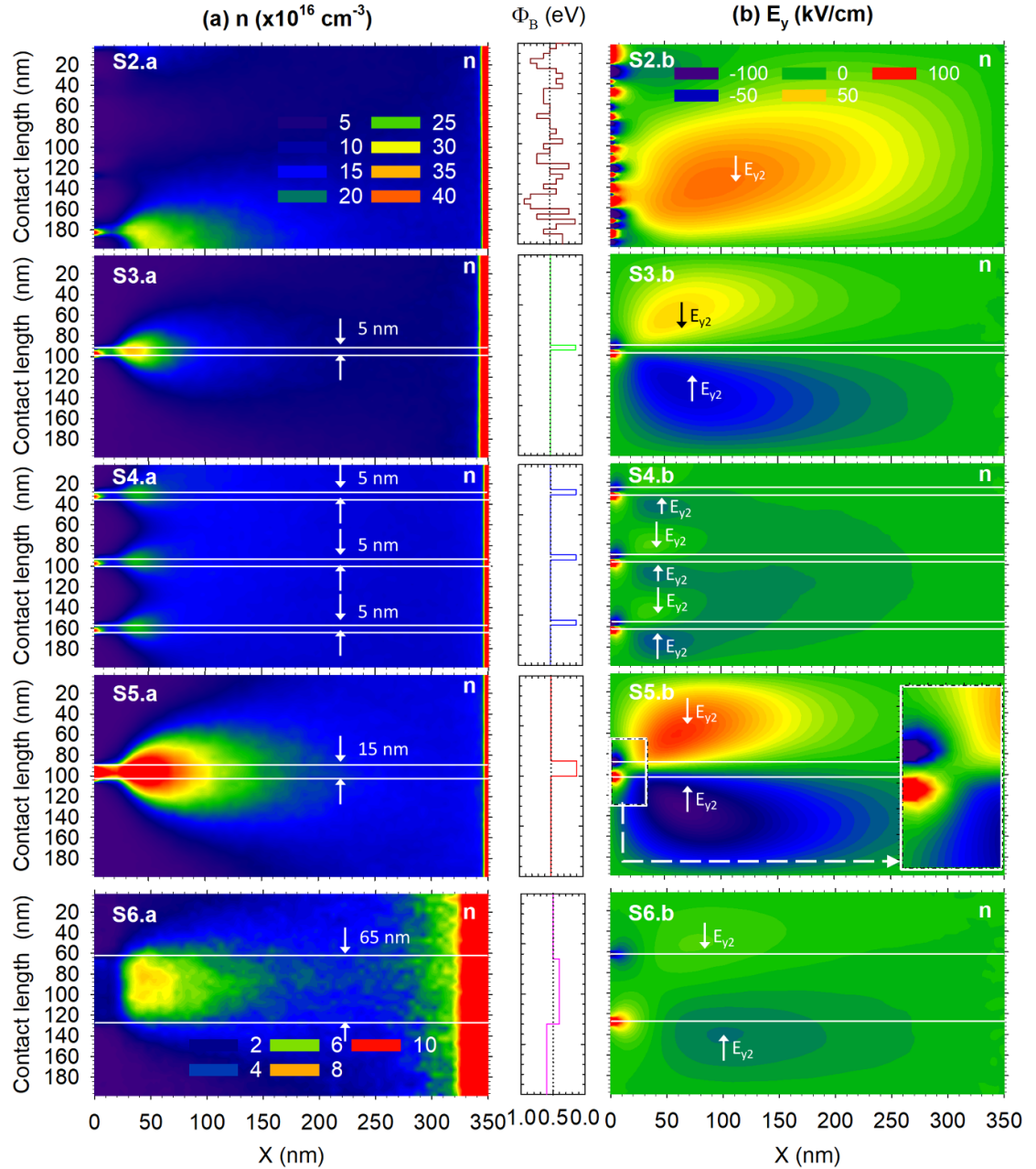


Figure 4. 2D profiles of (a) electron concentration (n) and (b) component of the electric field parallel to the Schottky contact (E_y) at $V = -30$ V for the simulated inhomogeneous ϕ_B distributions: S2, S3, S4, S5 and S6. Note that S6 electron concentration has different color scale. The inset in S5.b focuses on the electric field near the interface with the Schottky contact. The barrier profiles have been vertically plotted for clarity.

This explanation can be extrapolated to the other barrier profiles S6 and S2 (figures 4(S6(a)), (S6(b)) and (S2(a)), (S2(b))). The low-barrier regions exhibit higher electron concentration due to higher tunneling probability and therefore higher tunneling injection. The electric field inside the semiconductor helps to expand the carriers into the other high-barrier regions, being these two-dimensional effects responsible for the higher current, in particular in S6 with respect to the addition of three non-interacting regions with the same contact length and ϕ_B values.

Once the reverse current has been studied, figure 5 shows (a) the inverse of the squared values of the simulated capacitance as a function of the reverse applied voltage, and (b) the forward I - V characteristics for the different structures considered. With respect to the current characteristics, S5 is also the barrier profile that provides the highest value of the current and the homogeneous barrier S1 the one with less current, as it happened for reverse bias. This is because the low-barrier regions enhance, in this case,

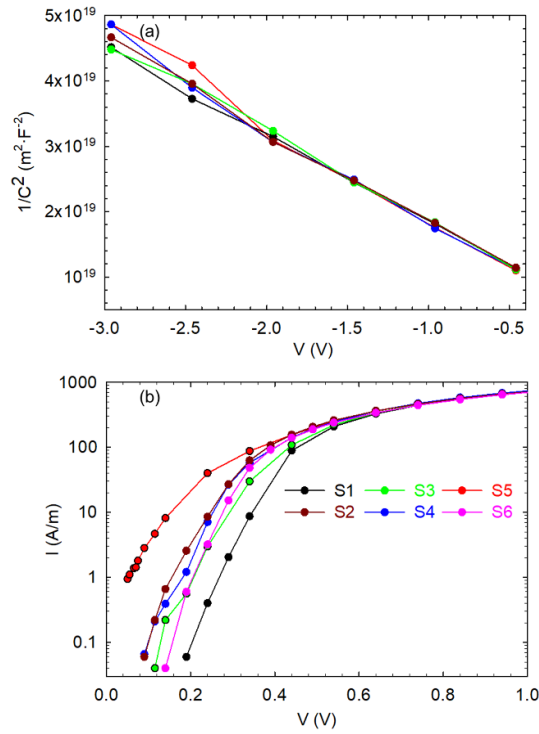
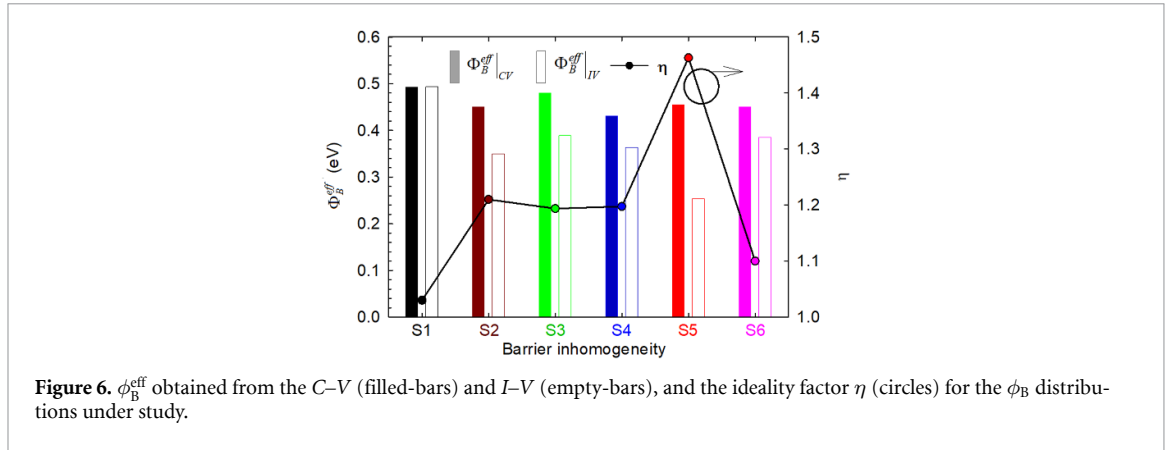


Figure 5. (a) Inverse of the squared capacitance per unit length as a function of the reverse voltage and (b) forward-bias I - V curves in logarithmic scale for a SBD with a homogeneous barrier of 0.5 V (S1) and (S2), (S3), (S4), (S5) and (S6) distributions.

the thermionic emission and provide more current. At this point, it is interesting to remark that similar reasoning related to 2D effects like those used for the reverse current can explain the difference among the forward I - V curves for some distributions, such as S4 and S5. For applied voltages higher than the barrier heights, the series resistance of the device dominates the I - V characteristics and, therefore, similar values of saturation current are reached for the different barrier profiles considered.

However, the C - V characteristics show very close values for the different inhomogeneous and homogeneous distributions. For the capacitance, full depletion still holds for low reverse bias, since the concentration of injected free carriers is negligible in comparison with the fixed charges in the depletion region. Therefore, the injection of carriers does not play a key role in the capacitance characteristics and the main influence is given by the value of the V_{bi} of the device according to (4). Although an inhomogeneous ϕ_B along the SC produces locally different values of V_{bi} and therefore the depletion region inside the semiconductor could be larger or shorter, the fact is that this local variation does not strongly affect the global magnitude of V_{bi} for the analyzed structures. In practice, it is almost negligible for low-barrier regions surrounded by a high ϕ_B background and is also nearly balanced in the device with the Gaussian distribution. This occurs because, at a distance of a few Debye lengths (L_D) inside the semiconductor from the SC, the potential profile is similar for the low-barrier and the high-barrier background regions, and thus the depletion region reaches approximately the same length. Consequently, the inhomogeneous ϕ_B along the contact importantly affects the current characteristics but has less impact on the capacitance ones.

In this sense, figure 6 shows the ϕ_B^{eff} obtained for the different barrier profile distributions extracted from the C - V and I - V characteristics of figures 5(a) and (b), fitting them to (1) and (4), respectively. From the fitting of the I - V curves, R_s values between $0.9 - 1.7 \text{ m}\Omega \cdot \text{m}^{-1}$ were extracted. From the C - V characteristics, we obtained a $\phi_B^{\text{eff}}|_{CV}$ (figure 6 (filled-bars)) with similar values for all the distributions, thus slightly affected by barrier inhomogeneities. On the other hand, the values of $\phi_B^{\text{eff}}|_{IV}$ calculated from I - V plots (figure 6 (empty-bars)) exhibit a wide dispersion. In addition, for the homogeneous device $\phi_B^{\text{eff}}|_{CV} \sim \phi_B^{\text{eff}}|_{IV}$, but this is not the case for the inhomogeneous SC distributions, where both effective values present significant differences. Results are in agreement with the behavior found in experimental devices, where there is also a discrepancy between the ϕ_B^{eff} values obtained from C - V and I - V measurements, being usually higher than the ones from the C - V [5]. From our simulations, we can assert that this contrast between the ϕ_B^{eff} values obtained from the capacitance and the current characteristics can be



justified as previously. Note that, when considering a Gaussian distribution (case S2), $\phi_B^{\text{eff}}|_{CV}$ corresponds to around the mean value of the barrier height [16], while $\phi_B^{\text{eff}}|_{IV}$ takes lower values than in cases S3, S4 and S6 because of the smaller local ϕ_B appearing in the barrier profile. The current is strongly enhanced by the regions of barriers with lower heights but, for the C - V curves, it is the higher barrier that dominates by pinching-off of the low-barrier regions under reverse bias [3].

If we analyze now the value of the ideality factor η for all the structures, also plotted in figure 6(circles), it is observed that the values differ from 1 when inhomogeneities appear in the SC, and the higher the difference between $\phi_B^{\text{eff}}|_{CV}$ and $\phi_B^{\text{eff}}|_{IV}$ the higher the value of η . This is a typical effect linked to inhomogeneities also in other technologies [29, 30]. On the other hand, the influence of the regions with a low barrier height depends not only on the width and value of the barrier but also on the location, because the 2D effects can be very decisive: S4 and S5 have very different η (1.20 and 1.46), although both structures globally have 15 nm of 0.1 eV in ϕ_B .

4. Conclusions

A deep microscopic study of the physical behavior of GaN SBDs with inhomogeneous ϕ_B profile has been performed by means of MC simulations. The analysis includes the electron concentration and vertical electric field profiles as well as the influence on the I - V and C - V characteristics for different realistic and more ideal ϕ_B distributions. The 2D reorganization of the tunneled electrons is connected to the increase observed in the reverse current, which mostly depends on the value and area of the low-barrier regions, but also on its relative location. The tunneled electrons through the low-barrier zones tend to move towards the surrounding low populated regions inside the semiconductor. When several low-barrier regions are close, the 2D carrier reorganization interferes with the global tunnel current. 2D effects emerge as an important aspect to be taken into account for the interpretation of the low values of ϕ_B^{eff} and the high values of η usually obtained in experimental devices from the I - V curves.

The ϕ_B^{eff} obtained from the C - V characteristics remains nearly constant due to the small influence of the low-barrier regions on the capacitance, being the high-barrier regions the ones that dominate by pinching-off the value of the low-barrier.

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

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